

Assignment 5: IAM & Logging

COMS/CPRE 4590X/5590 Spring 2026

Total Credits: 20 points

Due: Wednesday, April 8

1. Lab 1: Fine-Grained IAM and Role-Based Access (12 pts)

Part A: Fine-Grained Policy

1. Create an S3 bucket named **securebucketx** (you can replace x with any string).
2. Create a policy **P1** with:
 - Allow listing buckets
 - Allow **GetObject** only for objects inside bucket **securebucketx**
 - Explicitly deny access to all other objects in S3
 - Allow read-only EC2 access (**Describe***)
3. Create IAM user **AuditorUser** and attach policy P1.
4. Log in as AuditorUser and verify:
 - Can list bucket
 - Can access objects in **securebucketx**
 - Cannot access other objects
 - Can List EC2 instances
 - Cannot start/stop EC2 instances
5. If any of the above cannot be verified, find out how to fix the problem.

Part B: Role-Based Access

6. Create role **AppRole**:
 - Permission: Policy **P1**
 - Trust policy: allows EC2 service to assume the role
7. Launch an EC2 instance with AppRole attached.
8. From the EC2 instance, using CLI, verify the following:
 - Can list bucket

- Can access objects in **securebucketx**
- Cannot access other objects
- Can List EC2 instances
- Cannot start/stop EC2 instances

9. If any of the above cannot be verified, find out how to fix the problem.

Submission: snapshots of the following items.

- IAM → Policies → P1 → JSON view [2pt]
- S3 → securebucketx → Permissions tab [1pt]
- IAM → Users → AuditorUser → Permissions tab [1pt]
- S3 access success and failure screenshots (before and after fixing bugs) [2pts]
- EC2 instance list (DescribeInstances) [1pt]
- IAM → Roles → AppRole
 - Permissions tab [1pt]
 - Trust relationships tab [1pt]
- EC2 CLI outputs in S3 access success and failure screenshots (before and after fixing bugs) [2pts]
- EC2 CLI outputs in EC2 instance list [1pt]

2. Lab 2: VPC Flow Logs and Traffic Analysis (8 pts)

This problem focuses on **network-level visibility** and the **IAM configuration** required for **cross-service logging**.

Overview

You will:

- Launch an EC2 instance running a web server
- Configure VPC Flow Logs
- Create an IAM role to allow logs to be delivered to CloudWatch
- Generate both allowed and rejected traffic
- Analyze the resulting logs

Step 1: Launch EC2 Instance

1. Go to EC2 → Instances → Launch Instance
2. Use default VPC and a default subnet
3. Ensure the instance has a public IP

Security Group:

- Allow HTTP (port 80) from anywhere
- (Optional) Allow SSH (port 22) for access

Run a simple HTTP server:

```
sudo python3 -m http.server 80
```

Step 2: Create CloudWatch Log Group

1. Go to CloudWatch → Log groups
2. Click “Create log group”
3. Name: `defaultVPCLogs`

Step 3: Create IAM Role for Flow Logs

Goal: Allow VPC Flow Logs to write to CloudWatch.

1. Go to IAM → Roles → Create role
2. Select:
 - Trusted entity: AWS service

- Use case: VPC

Attach Permissions:

Create or attach a policy that allows:

- logs:CreateLogGroup
- logs:CreateLogStream
- logs:PutLogEvents

Verify Trust Policy (IMPORTANT):

```
{
  "Effect": "Allow",
  "Principal": {
    "Service": "vpc-flow-logs.amazonaws.com"
  },
  "Action": "sts:AssumeRole"
}
```

Name the role:

```
VPCFlowLogsRole
```

Step 4: Enable VPC Flow Logs

1. Go to VPC → Your VPCs
2. Select your VPC
3. Go to the **Flow Logs** tab
4. Click “Create flow log”

Configuration:

- Filter: ALL (capture ACCEPT and REJECT)
- Destination: CloudWatch Logs
- Log group: defaultVPCLogs
- IAM role: VPCFlowLogsRole

Step 5: Generate Traffic

From your local machine:

Allowed traffic (HTTP):

```
curl http://<EC2-public-IP>
```

Rejected traffic:

- Attempt connection to a closed port (e.g., port 81), OR
- Modify security group to block a port and test access

Step 6: View Logs in CloudWatch

1. Go to CloudWatch → Log groups
2. Select **defaultVPCLogs**
3. Open a log stream

Identify:

- ACCEPT entries (successful traffic)
- REJECT entries (blocked traffic)

Submission

- IAM → Roles → VPCFlowLogsRole:
 - Permissions tab [1pt]
 - Trust relationships tab [1pt]
- VPC → Your VPC → Flow Logs tab [1pt]
- CloudWatch → Log groups → defaultVPCLogs [1pt]
 - Log streams list
 - One opened log stream (partial)
- Screenshot of
 - One ACCEPT record [1pt]
 - One REJECT record [1pt]
- Explanation
 - Why the above traffic was accepted [1pt]
 - Why the above traffic was rejected [1pt]